From One Chart to a Full Financial Model

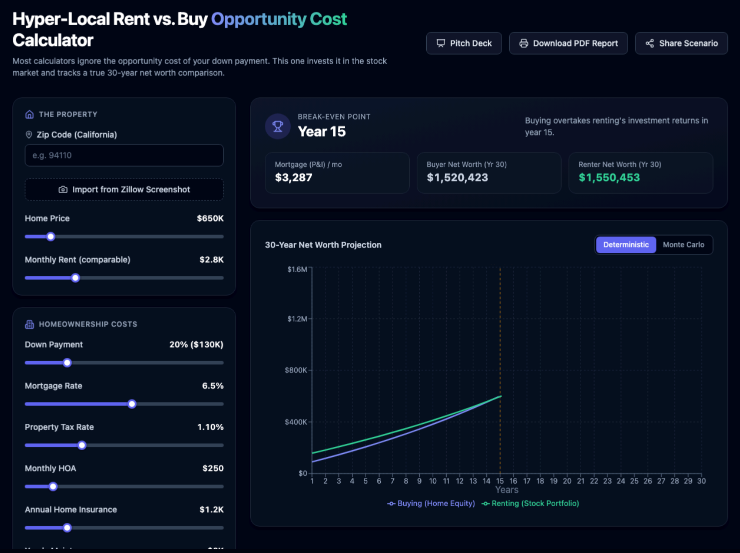
How the Rent vs. Buy Calculator Evolved

Every capability on the following slides started as a question — not a spec.

Where We Started

Most rent-vs-buy calculators only compare monthly payments. Build one that also asks what the down payment could have earned elsewhere.

- A 30-year net worth comparison, renter vs. buyer
- One clear answer: the **break-even year**



THE QUESTION

> *"How can we make this feel local, not generic?"*

Zip-code-aware defaults

Type a zip code, get real market numbers instantly — home price, rent, and property tax rate — instead of guessing at three unfamiliar sliders.

THE QUESTION

"What costs are we missing?"

Insurance and maintenance, modeled properly

Home insurance and maintenance don't inflate at the same pace as rent, so they got their own sliders and their own inflation rates — not lumped together as one vague "cost of owning" number.

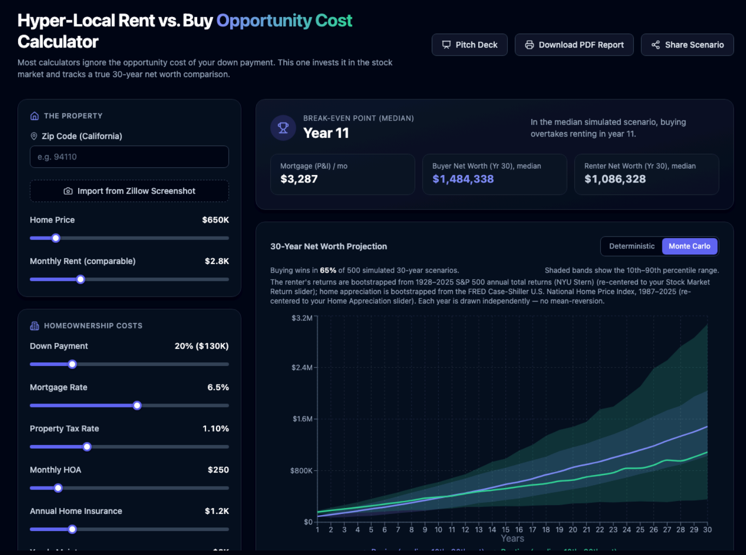
The section was renamed from "Mortgage & Costs" to **"Homeownership Costs"** to match what it actually covered by this point.

THE QUESTION

> *"A single guess isn't good enough — what about risk?"*

A second view: Monte Carlo simulation

Markets don't move in a straight line. A toggleable second view runs 500 simulated 30-year outcomes and shows a range, not false precision.



THE QUESTION

> *"Can that simulation be grounded in real history, not a guess?"*

Real market history, not invented statistics

The simulation originally sampled from a made-up statistical curve. It now resamples from what actually happened:

- Real S&P 500 annual returns, **1928–2025**
- The real Case-Shiller national home price index, **1987–2025**

Every simulated year is a year that genuinely occurred, just reshuffled.

THE QUESTION

"What about taxes — isn't that the biggest lever?"

Modeling what actually shows up on a tax return

- Mortgage interest and property tax deductions, weighed against the standard deduction
- Capital gains tax on the renter's investment gains
- The home-sale tax exclusion buyers get — $250K or $500K tax-free — which most calculators ignore entirely

THE QUESTION

"Taxes got complicated — can we make sense of it?"

Giving taxes their own home

Every tax-related input moved into one dedicated section. A state tax rate was added too, since California taxes investment gains as ordinary income — another detail most calculators miss.

THE QUESTION

"How do I share exactly what I'm looking at?"

One click, one link

Every input and every setting — encoded into a single URL. No accounts, no saved files, no re-explaining your numbers to someone else.

THE QUESTION

"Do I have to type in every number by hand?"

Upload a screenshot instead

Drop in a screenshot of a listing and the key numbers get pulled out automatically. Nothing is applied silently — every extracted value is shown for review and confirmation first.

THE QUESTION

"Can I take this analysis with me?"

A real report, one click away

Your exact scenario — every chart, every input, every assumption — turns into a clean, shareable document instead of living only on screen.

THE QUESTION

"What if I don't want to bet on the stock market?"

Choose your own investment

Stocks, Treasury bonds, or gold — each with its own real historical data behind it. Same rigor, whichever path you're actually weighing.

Every Feature Started With a Question

None of this was planned upfront. Each capability came from asking **"can we also..."** — and the model just kept getting more honest.

onevvish.github.io/rentVsbuy